Lección 1

Nos conectamos a Debian con el comando:

→ ssh -p 22022 ignaciordt@192.168.56.103

→ cat /proc/mdstat para ver el estado de los dispositivos RAID gestionados por Linux

```
ignaciordt@debian-ise:~$ cat /proc/mdstat
Personalities : [raid1] [raid0] [raid6] [raid5] [raid4] [raid10]
md0 : active raid1 sdb1[1] sda1[0]
      486400 blocks super 1.2 [2/2] [UU]

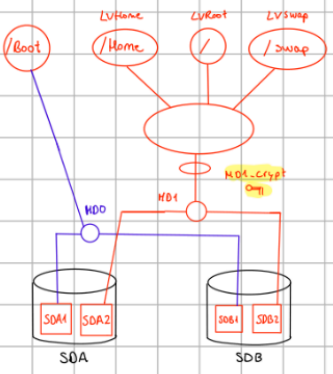
md1 : active raid1 sda2[0] sdb2[1]
      9987072 blocks super 1.2 [2/2] [UU]

unused devices: <none>
```

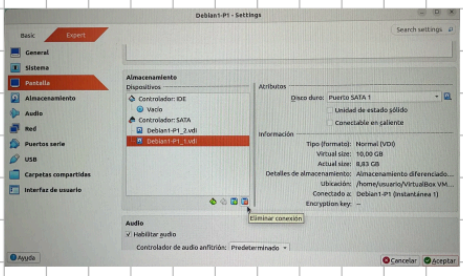
→ El RAID tiene 2 componentes activos de los 2 esperados.

→ lsblk

```
ignaciordt@debian-ise:~$ lsblk
NAME                          MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda                             8:0    0   10G  0 disk
├─sda1                          8:1    0  476M  0 part
│ └─md0                         9:0    0  475M  0 raid1 /boot
└─sda2                          8:2    0  9,5G  0 part
  └─md1                         9:1    0  9,5G  0 raid1
    ├─vg_raid1-root           253:0    0  6,5G  0 lvm
    │ └─vg_raid1-root_crypt   253:3    0  6,5G  0 crypt /
    ├─vg_raid1-home           253:1    0  980M  0 lvm
    │ └─vg_raid1-home_crypt   253:5    0  964M  0 crypt /home
    └─vg_raid1-swap           253:2    0  952M  0 lvm
      └─vg_raid1-swap_crypt   253:4    0  936M  0 crypt [SWAP]
sdb                             8:16   0   10G  0 disk
├─sdb1                          8:17   0  476M  0 part
│ └─md0                         9:0    0  475M  0 raid1 /boot
└─sdb2                          8:18   0  9,5G  0 part
  └─md1                         9:1    0  9,5G  0 raid1
    ├─vg_raid1-root           253:0    0  6,5G  0 lvm
    │ └─vg_raid1-root_crypt   253:3    0  6,5G  0 crypt /
    ├─vg_raid1-home           253:1    0  980M  0 lvm
    │ └─vg_raid1-home_crypt   253:5    0  964M  0 crypt /home
    └─vg_raid1-swap           253:2    0  952M  0 lvm
      └─vg_raid1-swap_crypt   253:4    0  936M  0 crypt [SWAP]
sr0                            11:0    1 1024M  1 rom
ignaciordt@debian-ise:~$
```



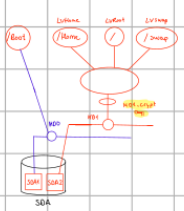
→ Ahora procedemos a quitar sdb



→ lsblk y vemos que se ha borrado correctamente

```
ignaciordt@debian-ise:~$ lsblk
NAME                          MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda                             8:0    0   10G  0 disk
├─sda1                          8:1    0  476M  0 part
│ └─md0                         9:0    0  475M  0 raid1 /boot
└─sda2                          8:2    0  9,5G  0 part
  └─md1                         9:1    0  9,5G  0 raid1
    ├─vg_raid1-root           253:0    0  6,5G  0 lvm
    │ └─vg_raid1-root_crypt   253:3    0  6,5G  0 crypt /
    ├─vg_raid1-home           253:1    0  980M  0 lvm
    │ └─vg_raid1-home_crypt   253:5    0  964M  0 crypt /home
    └─vg_raid1-swap           253:2    0  952M  0 lvm
      └─vg_raid1-swap_crypt   253:4    0  936M  0 crypt [SWAP]
sr0                            11:0    1 1024M  1 rom
ignaciordt@debian-ise:~$
```

→ cat /proc/mdstat para ver el estado de los dispositivos RAID gestionados por Linux




```
ignaciordt@debian-ise:~$ cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid1 sda1[0]
      486400 blocks super 1.2 [2/1] [U_]

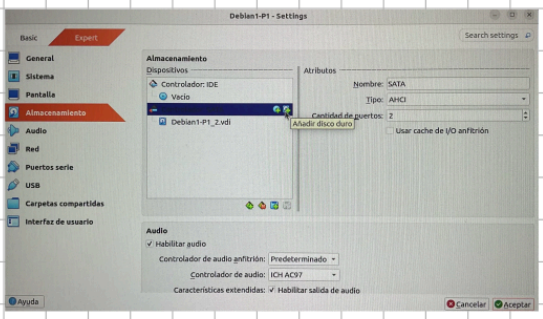
md1 : active raid1 sda2[0]
      9987072 blocks super 1.2 [2/1] [U_]

unused devices: <none>
ignaciordt@debian-ise:~$
```

→ El RAID tiene 1 componente activo de los 2 esperados.

→ Ahora creamos otro disco duro (Debian1-P1_nuevo.vdi)

→ lsblk




```
ignaciordt@debian-ise:~$ lsblk
NAME                      MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda                         8:0    0   10G  0 disk
├─sda1                      8:1    0  476M  0 part
│ └─md0                     9:0    0  475M  0 raid1 /boot
└─sda2                      8:2    0  9,5G  0 part
  └─md1                     9:1    0  9,5G  0 raid1
    ├─vg_raid1-root       253:0    0  6,5G  0 lvm
    │ └─vg_raid1-root_crypt 253:3  0  6,5G  0 crypt /
    ├─vg_raid1-home       253:1    0  980M  0 lvm
    │ └─vg_raid1-home_crypt 253:5  0  964M  0 crypt /home
    └─vg_raid1-swap       253:2    0  952M  0 lvm
      └─vg_raid1-swap_crypt 253:4  0  936M  0 crypt [SWAP]
sdb                         8:16   0   20G  0 disk
sr0                        11:0    1 1024M  1 rom
```

→ su -

→ fdisk /dev/sdb n, p, 1, 2048, 976895 { t, 1, fd } w
n, p, 2, 976896, 20969471 { t, 2, fd }

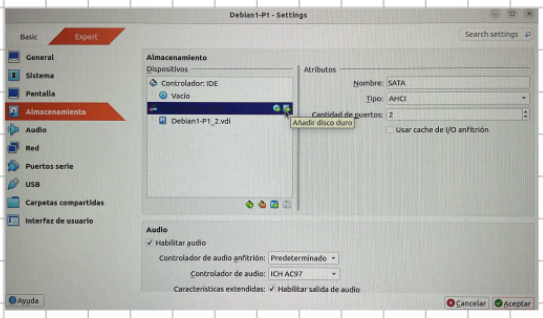
Preparamos la partición del nuevo disco.

→ lsblk

```
root@debian-ise:~# lsblk
NAME                      MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda                         8:0    0   10G  0 disk
├─sda1                      8:1    0  476M  0 part
│ └─md0                     9:0    0  475M  0 raid1 /boot
└─sda2                      8:2    0  9,5G  0 part
  └─md1                     9:1    0  9,5G  0 raid1
    ├─vg_raid1-root       253:0    0  6,5G  0 lvm
    │ └─vg_raid1-root_crypt 253:3  0  6,5G  0 crypt /
    ├─vg_raid1-home       253:1    0  980M  0 lvm
    │ └─vg_raid1-home_crypt 253:5  0  964M  0 crypt /home
    └─vg_raid1-swap       253:2    0  952M  0 lvm
      └─vg_raid1-swap_crypt 253:4  0  936M  0 crypt [SWAP]
sdb                         8:16   0   20G  0 disk
├─sdb1                      8:17   0  476M  0 part
└─sdb2                      8:18   0  9,5G  0 part
sr0                        11:0    1 1024M  1 rom
root@debian-ise:~#
```

→ Ahora creamos otro disco duro (Debian1-P1_nuevo.vdi)

→ lsblk



ignaciordt@debian-ise:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 10G 0 disk
├─sda1 8:1 0 476M 0 part
│ └─md0 9:0 0 475M 0 raid1 /boot
└─sda2 8:2 0 9,5G 0 part
└─md1 9:1 0 9,5G 0 raid1
├─vg_raid1-root 253:0 0 6,5G 0 lvm
│ └─vg_raid1-root_crypt 253:3 0 6,5G 0 crypt /
├─vg_raid1-home 253:1 0 980M 0 lvm
│ └─vg_raid1-home_crypt 253:5 0 964M 0 crypt /home
└─vg_raid1-swap 253:2 0 952M 0 lvm
└─vg_raid1-swap_crypt 253:4 0 936M 0 crypt [SWAP]
sdb 8:16 0 20G 0 disk
sr0 11:0 1 1024M 1 rom

→ su -

→ fdisk /dev/sdb n, p, 1, 2048, 976895 { t, 1, fd } w
n, p, 2, 976896, 20969471 { t, 2, fd }

Preparamos la partición del nuevo disco.

→ lsblk

root@debian-ise:~# lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 10G 0 disk
├─sda1 8:1 0 476M 0 part
│ └─md0 9:0 0 475M 0 raid1 /boot
└─sda2 8:2 0 9,5G 0 part
└─md1 9:1 0 9,5G 0 raid1
├─vg_raid1-root 253:0 0 6,5G 0 lvm
│ └─vg_raid1-root_crypt 253:3 0 6,5G 0 crypt /
├─vg_raid1-home 253:1 0 980M 0 lvm
│ └─vg_raid1-home_crypt 253:5 0 964M 0 crypt /home
└─vg_raid1-swap 253:2 0 952M 0 lvm
└─vg_raid1-swap_crypt 253:4 0 936M 0 crypt [SWAP]
sdb 8:16 0 20G 0 disk
├─sdb1 8:17 0 476M 0 part
└─sdb2 8:18 0 9,5G 0 part
sr0 11:0 1 1024M 1 rom
root@debian-ise:~#

* → `mdadm --add /dev/md0 /dev/sdb1`

↳ Añadimos la partición nueva sdb1 al RAID1 md0 para recuperar la copia que perdimos.

* → `mdadm --add /dev/md1 /dev/sdb2`

↳ Añadimos la partición nueva sdb2 al RAID1 md1 para recuperar la copia que perdimos.

* → `watch -n 1 cat /proc/mdstat`

↳ Para ver en tiempo real cómo avanza la copia del RAID1 hasta que aparece [UU] → ¡ambos componentes están operativos!

```
root@debian-ise:~# mdadm --add /dev/md0 /dev/sdb1
mdadm: added /dev/sdb1
root@debian-ise:~# mdadm --add /dev/md1 /dev/sdb2
mdadm: added /dev/sdb2
root@debian-ise:~# watch -n 1 cat /proc/mdstat
root@debian-ise:~# lsblk
NAME                          MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda                             8:0    0   10G  0 disk
├─sda1                          8:1    0  476M  0 part
│ └─md0                         9:0    0  475M  0 raid1 /boot
└─sda2                          8:2    0  9,5G  0 part
  └─md1                         9:1    0  9,5G  0 raid1
    ├─vg_raid1-root           253:0    0  6,5G  0 lvm
    │ └─vg_raid1-root_crypt   253:3    0  6,5G  0 crypt /
    ├─vg_raid1-home           253:1    0  980M  0 lvm
    │ └─vg_raid1-home_crypt   253:5    0  964M  0 crypt /home
    └─vg_raid1-swap           253:2    0  952M  0 lvm
      └─vg_raid1-swap_crypt   253:4    0  936M  0 crypt [SWAP]
sdb                             8:16   0   10G  0 disk
├─sdb1                          8:17   0  476M  0 part
│ └─md0                         9:0    0  475M  0 raid1 /boot
└─sdb2                          8:18   0  9,5G  0 part
  └─md1                         9:1    0  9,5G  0 raid1
    ├─vg_raid1-root           253:0    0  6,5G  0 lvm
    │ └─vg_raid1-root_crypt   253:3    0  6,5G  0 crypt /
    ├─vg_raid1-home           253:1    0  980M  0 lvm
    │ └─vg_raid1-home_crypt   253:5    0  964M  0 crypt /home
    └─vg_raid1-swap           253:2    0  952M  0 lvm
      └─vg_raid1-swap_crypt   253:4    0  936M  0 crypt [SWAP]
sr0                            11:0    1 1024M  1 rom
root@debian-ise:~#
```

```
Cada 1,0s: cat /proc/mdstat                    debian-ise: Sun Nov  1 21:02:06 2026

Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid1 sdb1[2] sda1[0]
      486400 blocks super 1.2 [2/2] [UU]

md1 : active raid1 sdb2[2] sda2[0]
      9987072 blocks super 1.2 [2/1] [U_]
      [======>.............]  recovery = 34.9% (3494720/9987072) finish=0.5min
speed=205571K/sec

unused devices: <none>
```

Vemos como se está copiando

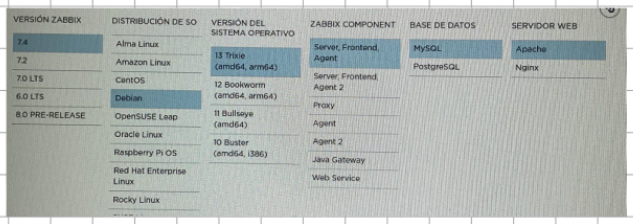
```
ignaciordt@debian-ise: ~
Cada 1,0s: cat /proc/mdstat                    debian-ise: Sun Nov  1 21:02:53 2026

Personalities : [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid1 sdb1[2] sda1[0]
      486400 blocks super 1.2 [2/2] [UU]

md1 : active raid1 sdb2[2] sda2[0]
      9987072 blocks super 1.2 [2/2] [UU]

unused devices: <none>
```

Instalación



- **Instalación del Repositorio:** Se descargó e instaló el paquete del repositorio oficial de Zabbix y se actualizaron las fuentes de apt.

```
# wget https://repo.zabbix.com/zabbix/7.4/release/debian/pool/main/z/zabbix-release/zabbix-release_latest_7.4+debian
# dpkg -i zabbix-release_latest_7.4+debian13_all.deb
# apt update
```

→ Todo se ejecuta en Debian13.

- **Instalación de Servidor, Interfaz y agente de Zabbix:**

```
# apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent
```

→ En root, hacemos `mysql -u root -p`

```
MariaDB [(none)]> create database zabbix character set utf8mb4 collate utf8mb4_bin;
Query OK, 1 row affected (0,002 sec)

MariaDB [(none)]> create user zabbix@localhost identified by 'password';
Query OK, 0 rows affected (0,036 sec)

MariaDB [(none)]> grant all privileges on zabbix.* to zabbix@localhost;
Query OK, 0 rows affected (0,030 sec)

MariaDB [(none)]> set global log_bin_trust_function_creators = 1;
Query OK, 0 rows affected (0,001 sec)

MariaDB [(none)]> quit;
```

Dentro de Mariadb hacemos esos comandos

→ Importamos el esquema inicial con la base de datos usando zcat.

```
# zcat /usr/share/zabbix/sql-scripts/mysql/server.sql.gz | mysql --default-character-set=utf8mb4 -uzabbix -p zabbix
```

→ Deshabilitamos la opción de log_bin_trust_function_creators

```
MariaDB [(none)]> set global log_bin_trust_function_creators = 0;
Query OK, 0 rows affected (0,001 sec)

MariaDB [(none)]> quit;
Bye
```

Configuración

→ su -

→ nano /etc/zabbix/zabbix_server.conf

↳ Ponemos la contraseña.

```
GNU nano 8.4                    /etc/zabbix/zabbix_server.conf *
# DBUser=

DBUser=zabbix

### Option: DBPassword
#       Database password.
#       Comment this line if no password is used.
#
# Mandatory: no
# Default:
DBPassword=practicas,ise

### Option: DBSocket
#       Path to MySQL socket.
```

→ systemctl restart zabbix-server zabbix-agent apache2 → Reiniciamos los cambios realizados.

→ systemctl status zabbix-server

↳ Para comprobar el estado del servidor.

```
root@debian-ise:~# systemctl restart zabbix-server zabbix-agent apache2
root@debian-ise:~# systemctl status zabbix-server
● zabbix-server.service - Zabbix Server
     Loaded: loaded (/usr/lib/systemd/system/zabbix-server.service; disabled; p
     Active: active (running) since Mon 2026-11-02 11:46:23 CET; 20s ago
  Invocation: 9a5d949d9416467095 1c6398a48e74c2
    Process: 1510 ExecStart=/usr/sbin/zabbix_server -c $CONFFILE (code=exited,
   Main PID: 1518 (zabbix_server)
      Tasks: 77 (limit: 2297)
     Memory: 60.7M (peak: 62M)
        CPU: 276ms
     CGroup: /system.slice/zabbix-server.service
             ├─1518 /usr/sbin/zabbix_server -c /etc/zabbix/zabbix_server.conf
```

→ systemctl enable zabbix-server zabbix-agent apache2

→ Para que se inicie automáticamente cada vez que arranquemos Debian.

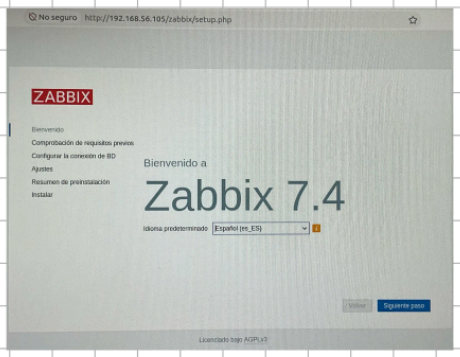
¡Error que me ha ocurrido pero sirve para ayuda!

→ Pongo en el navegador http://192.168.56.105/zabbix y se me queda cargando.

Porque teníamos configurado el cortafuegos UFW (porque estamos en Debian) para permitir SSH por el puerto 22022, pero no hemos permitido HTTP por el puerto 80.

→ ufw allow 80/tcp

→ Ahora buscamos http://192.168.56.105/zabbix.

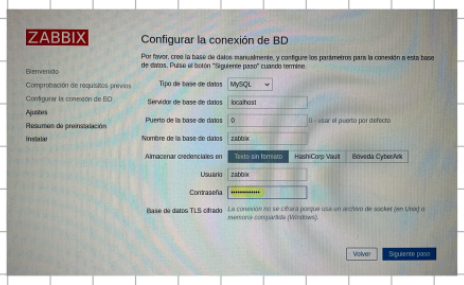


→ Comprobamos que está todo OK.

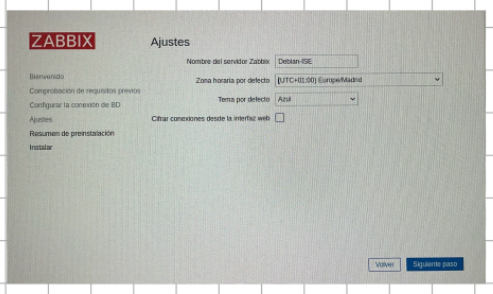


La que pusimos en zabbix_server.conf

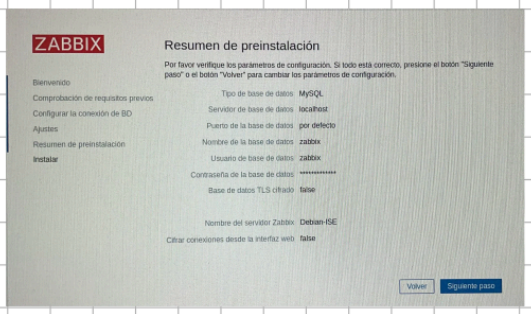
→ Dejamos todo como está y ponemos la contraseña practicas,ise.

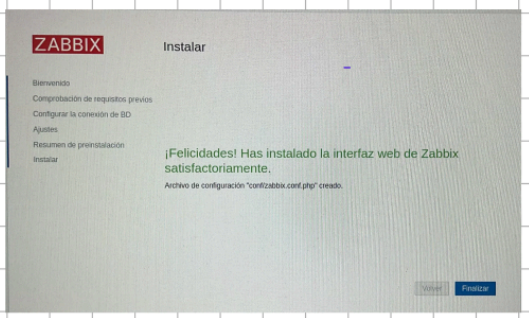


→ Ponemos el nombre y ajustamos los cambios.

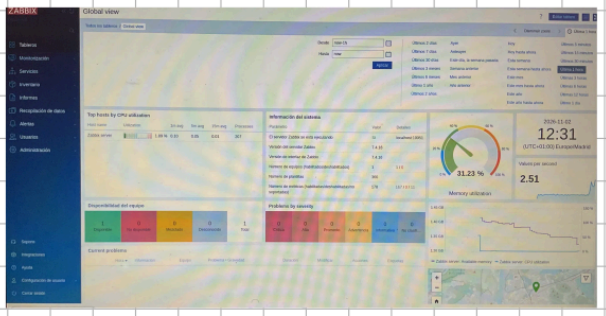


→ Confirmamos los datos de la configuración.





→ Entramos con Admin (usuario) y ~~Zabbix~~ (contraseña)



→ ¡Ya tenemos a Zabbix 7.4 instalado correctamente en Debian 13!

Instalación

→ ssh ignaciordt@192.168.56.102 -p 22022 para conectarme a almaRaid desde la terminal.

En almaRAID

```
[ignaciordt@vbox ~]$ sudo rpm -Uvh https://repo.zabbix.com/zabbix/7.4/release/rhel/10/noarch/zabbix-release-latest-7.4.el10.noarch.rpm
Retrieving https://repo.zabbix.com/zabbix/7.4/release/rhel/10/noarch/zabbix-release-latest-7.4.el10.noarch.rpm
warning: /var/tmp/rpm-tmp.JqXOek: Header V4 RSA/SHA512 Signature, key ID b5333005: NOKEY
Verifying...                          ################################# [100%]
Preparing...                          ################################# [100%]
Updating / installing...
   1:zabbix-release-7.4-4.el10        ################################# [100%]
```

```
[ignaciordt@vbox ~]$ sudo dnf install zabbix-agent
Last metadata expiration check: 0:00:09 ago on Fri Oct 16 03:38:23 2026.
Dependencies resolved.
================================================================================
 Package             Architecture  Version                     Repository     Size
================================================================================
Installing:
 zabbix-agent        x86_64        7.4.16-release2.el10        zabbix        657 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 657 k
Installed size: 2.9 M
Is this ok [y/N]: y
```

→ Instalamos Zabbix Agent, el programa que permite al servidor Zabbix de Debian recopilar información de Almalinux

→ sudo nano /etc/zabbix/zabbix_agentd.conf

```
GNU nano 8.1                    /etc/zabbix/zabbix_agentd.conf                    Modified
# LogRemoteCommands=0

##### Passive checks related

### Option: Server
#       List of comma delimited IP addresses, optionally in CIDR notation, or DNS n
#       Incoming connections will be accepted only from the hosts listed here.
#       If IPv6 support is enabled then '127.0.0.1', '::127.0.0.1', '::ffff:127.0.0
#       and '::/0' will allow any IPv4 or IPv6 address.
#       '0.0.0.0/0' can be used to allow any IPv4 address.
#       Example: Server=127.0.0.1,192.168.1.0/24,::1,2001:db8::/32,zabbix.example.c
#
# Mandatory: yes, if StartAgents is not explicitly set to 0
# Default:
# Server=

Server=192.168.56.105

### Option: ListenPort
#       Agent will listen on this port for connections from the server.

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

DEBIAN		ALMALINUX
192.168.56.105		192.168.56.102
Zabbix Server (pregunta)	→	Zabbix Agent (responde)

Solo permito que el servidor Zabbix de Debian consulte para obtener las métricas en Almalinux.

```
# Mandatory: no
# Default:
# ServerActive=

ServerActive=192.168.56.105

### Option: Hostname
#       List of comma delimited unique, case
#       Required for active checks and must
#       Value is acquired from HostnameItem
#
# Mandatory: no
```

- `Server` → Indica qué servidor puede **preguntar** al agente por sus métricas (comprobaciones pasivas).
- `ServerActive` → Indica a qué servidor debe **enviar** el agente los datos de las comprobaciones activas.

En ambos casos utilizamos la IP de **Debian**, porque allí está nuestro Zabbix Server.

```
# Mandatory: no
# Default:
# Hostname=

Hostname=AlmaLinux-ISE
```

Idea clave: el parámetro `Hostname` identifica al agente y debe coincidir con el nombre técnico del host configurado en Zabbix, especialmente para las comprobaciones activas.

Iniciamos, comprobamos y habilitamos el servicio zabbix-agent.

→ `sudo systemctl start zabbix-agent`

→ `sudo systemctl status zabbix-agent`

→ `sudo systemctl enable zabbix-agent`

→ Abrimos el puerto 10050, lo permitimos SOLO desde Debian.

```
[ignaclordt@vbox ~]$ sudo firewall-cmd --permanent --add-rich-rule='rule family="ipv
4" source address="192.168.56.105" port port="10050" protocol="tcp" accept'
success
[ignaclordt@vbox ~]$ sudo firewall-cmd --reload
success
[ignaclordt@vbox ~]$
```

```
ignaclordt@192.168.56.105's password:
Linux debian-ise 6.12.107-debi3-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (
2026-08-29) x86_64

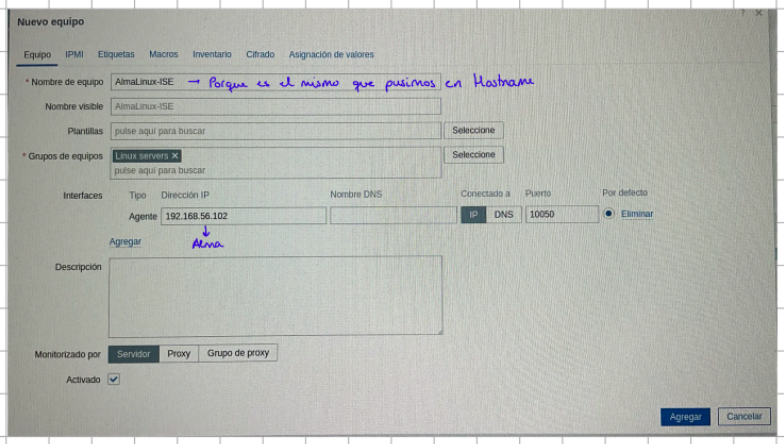
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Nov  1 22:22:19 2026 from 192.168.56.1
ignaclordt@debian-ise: $ nc -vz -w 3 192.168.56.102 10050
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 10050 (zabbix-agent) open
```

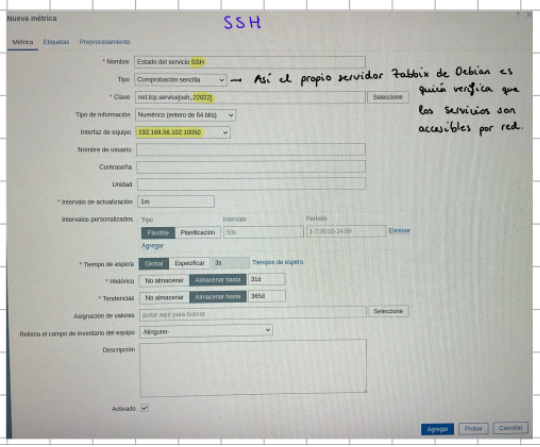
Comprobamos que desde Debian podemos conectarnos a zabbix-agent.

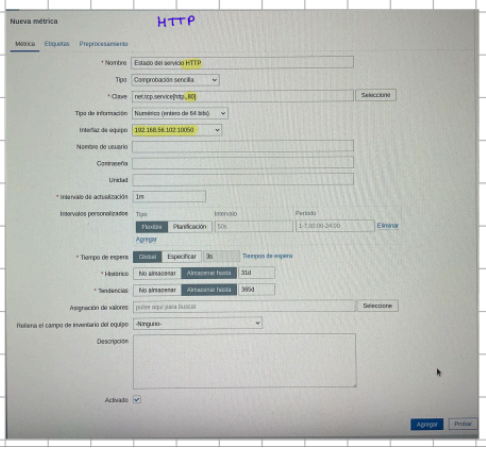
Creación de Host e Items (SSH y HTTP)

→ Desde http://192.168.56.105/zabbix Vamos a → Recopilación de Datos → Equipos → Crear Equipo

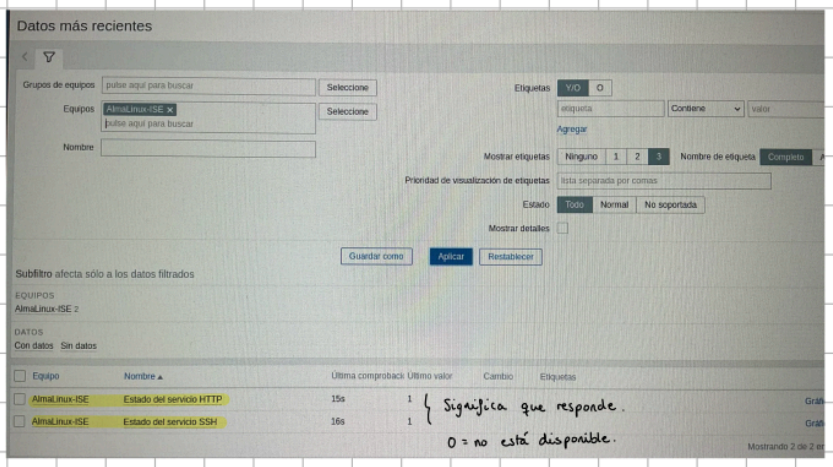


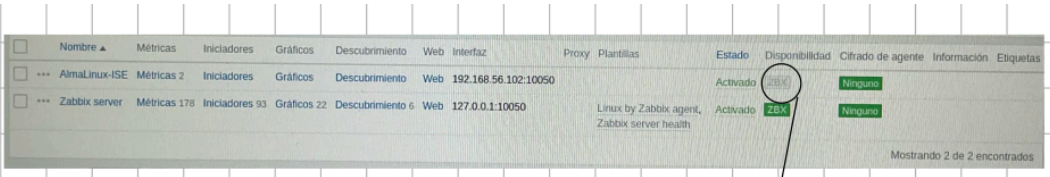
→ En Equipos → Métricas → Crear Métrica



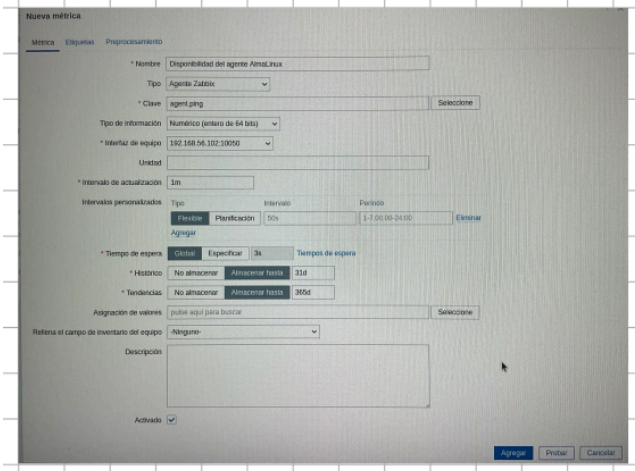


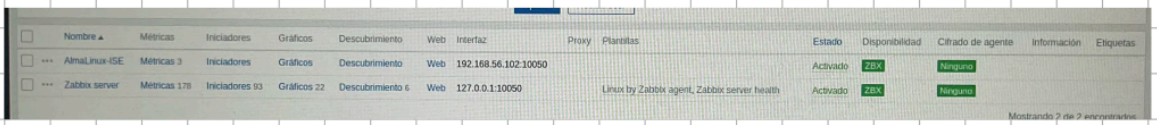
Vamos a Monitorización → Datos recientes





Se tiene que poner en verde.





Esto significa que:
- → Debian (Zabbix Server): Zabbix monitoriza correctamente el propio servidor.
- → AlmaLinux-ISE: Zabbix Server se comunica correctamente con su agente.
- → SSH y HTTP de AlmaLinux ya comprobamos que ambos marcan 1.